MIL-T-52332A(ME)
4 January 1974

SUPERSEDING
MIL-T-52332(MO)
25 October 1963

MILITARY SPECIFICATION

TOOL, HORIZONTAL CONNECTING: ERECTION EQUIPMENT FOR BRIDGE, FLOATING, PNEUMATIC FLOAT, CLASS 60

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a horizontal connecting tool to be used as an erection aid in the construction of the class 60 floating bridge.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-P-386	- Pins, Cotter (Split).
FF-S-200	- Setscrews: Hexagon Socket and Spline Socket, Headless.
QQ-T-830	- Tube, Steel, (Carbon, Mechanical; Seamless and Welded).

FSC 5420

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:15Z
Check the source to verify that this is the current version before use.

RR-C-271	- Chains and Attachments, Welded, Weldless, and Roller Chain.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple Wall.
PPP-T-60	- Tape: Packaging, Waterproof.

Military

MIL-T-704	- Treatment and Painting of Materiel.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Testing.
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold-Application.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWINGS

ME

D7944-1 and -2	- Tool, Horizontal Connecting: Erection Equipment for Bridge, Floating Pneumatic Float, Class 60.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:15Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A36	- Structural Steel.
A304	- Alloy Steel Bars to End-Quench Hardenability Requirements.
A441	- High-Strength, Low-Alloy Structural Manganese Vanadium Steel.
A502	- Steel Structural Rivets.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY

A5.1	- Mild Steel Covered Arc-Welding Electrodes.
A5.18	- Mild Steel Electrodes for Gas Metal-Arc Welding.
B3.0	- Standard Qualification Procedure.
D1.1	- Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, 2501 Northwest Seventh Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street N.W., Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The horizontal connecting tool shall be as shown on Drawings D7944-1 and -2 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish a horizontal connecting tool for inspection within the time frame specified (see 6.2) to prove, prior to starting production, that his production methods will produce horizontal connecting tools that comply with the requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:15Z
Check the source to verify that this is the current version before use.

the supplier and shall be subject to all provisions of this specification.

3.3.1 Structural steel. Structural steel shall conform to ASTM A36.

3.3.2 Low-alloy structural steel. Low-alloy structural steel shall conform to ASTM A441.

3.4 Inserts. Inserts shall be made of steel conforming to ASTM A304, grade designation 4140H, with a Rockwell hardness of C45 to C50. The inserts shall not be case hardened. The teeth shall be machined and shaped.

3.5 Pins. Pins shall be fabricated of steel conforming to ASTM A304, grade designation 4140H, and shall be heat treated to a tensile strength of 125,000 psi.

3.6 Hydraulic ram. The hydraulic ram shall have a capacity of 10 tons with an internal pressure between 8,000 and 9,000 psi. The materials used in the hydraulic ram shall be compatible with fluid conforming to MIL-H-6083 or MIL-H-5606. When shipped, the hydraulic ram shall not be filled with hydraulic fluid, except as specified in 5.1.1 and 5.1.2.

3.7 Hydraulic couplings. A set of hydraulic couplings shall consist of a female section for attachment to the ram; male section for attachment to a hydraulic hose; swivel of female section; and dust caps attached to the male and female sections with steel chain conforming to RR-C-271, type II, class 3, weldless, sash chain, zinc coated. The coupling components shall be capable of withstanding a minimum of 10,000 psi internal working pressure without any evidence of leakage or failure. All coupling parts shall be of corrosion-resistant material or shall be rendered corrosion resistant by chromium or cadmium plating. The coupling shall be of such design that breaking the connection at the swivel will cause spring-loaded devices to seal the fluid passage in each half coupling. The spring-loaded devices shall open to permit the fluid flow when a connection is made. The exterior of the coupling shall be shaped to provide a wrench grip. The swivel shall turn freely by hand.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

3.8 Rivets. Rivets shall conform to ASTM A502, grade 1.

3.9 Chain. Chain shall conform to RR-C-271, type I, grade C, class 1.

3.10 Steel tubing. Steel tubing shall conform to QQ-T-830, seamless, hot rolled, round, composition 1025.

3.11 Set screws. Set screws shall conform to FF-S-200, type I, style 1.

3.12 Cotter pins. Cotter pins shall conform to FF-P-386, type B.

3.13 Welding electrodes. Welding electrodes shall conform to AWS A5.1 or A5.18, classification and size as required.

3.14 Identification marking. The horizontal connecting tool shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.15 Treatment and painting. The horizontal connecting tool, except for the couplings, pins, screws, and threads, shall be treated and painted in accordance with MIL-T-704, type B except that the color number designation shall be 34087 in accordance with FED. STD. No. 595. Pins, screws, and threads shall be coated with preservative conforming to MIL-C-16173, grade 1.

3.16 Workmanship.

3.16.1 Steel fabrication. Steel used in the fabrication of the horizontal connecting tool shall be free from laminations, kinks, and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and clipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting with a tip suitable for the thickness of metal may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. All bends shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. All burrs and rough edges shall be ground smooth.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:15Z
Check the source to verify that this is the currentversion before use.

3.16.2 Welders and welding.

3.16.2.1 Welders. Before assigning any welder or automatic welder equipment operator to welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder or operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society (B3.0).
Welding Qualification of the ASME Boiler and Pressure Vessel Code (Section IX).

3.16.2.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the American Welding Society publication "D1.1-Structural Welding Code". All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:15Z
Check the source to verify that this is the currentversion before use.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction horizontal connecting tool shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The preproduction horizontal connecting tool shall be tested as specified in 4.5.2. Failure to pass the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The horizontal connecting tools shall be examined as specified herein, for the following defects:

Major

101. Nonconformance to the dimensions shown on the drawings.
102. Components missing or not as specified.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:15Z
Check the source to verify that this is the currentversion before use.

103. Material not as specified.
104. Workmanship not as specified.
105. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.

Minor

201. Identification marking incorrect, missing, or illegible.
202. Treatment and painting not as specified.

4.5.2 Horizontal connecting tool test. Place the tube hook and the chain hook of the horizontal connecting tool over separate edges (approximately 28-1/2 inches apart) of an unyielding surface. Install a 10,000-psi-minimum pump to the horizontal connecting tool incorporating the coupling set and a 10,000-psi, minimum-capacity gage in the hydraulic line between the pump and tool. Pump hydraulic fluid into the tool until the gage in the line indicates the pressure required to produce 10 tons. Maintain this pressure on the gage for 5 minutes. Then release the pressure and disconnect the couplings at the swivel. Determine whether there is leakage through or around the seal. Connect the couplings again. Determine whether there is obstruction of flow. Evidence of any of the following shall constitute failure of this test:

(a) Failure or damage to the horizontal connecting tool.
(b) Any leakage from the couplings.
(c) Any leakage through or around the coupling seals when the couplings are disconnected.
(d) Any obstruction of hydraulic fluid flow when the couplings are connected.
(e) Any leakage of hydraulic fluid or evidence of damage to the hydraulic ram.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects:

106.	Materials and containers not as specified for level A or B.	Each incorrect material or container shall be considered one defect.
107.	Interior surfaces of the hydraulic ram not preserved as specified for level A or B.	
108.	Openings not sealed as specified for level A or B.	
109.	Contents not immobilized as specified for level A or B.	
110.	Box not waterproof sealed as specified for level A.	
111.	Quantities packed together exceed the weight or size limitations for level A or B.	
112.	Strapping not zinc coated for level A.	
113.	Marking missing, illegible, incorrect, or incomplete for level A, B, or C.	

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Interior surfaces of the hydraulic ram shall be coated with the hydraulic fluid recommended for operation and the excess fluid drained. Openings into the ram shall be sealed with tape conforming to PPP-T-60, type IV. The tool with couplings shall be packaged in a close fitting box conforming to PPP-B-636, W5c. The contents shall be immobilized within the box in a manner to prevent free movement or damage by the use of fiberboard inserts, pads, or cushions, fabricated from the same material as the box. The box shall be waterproof sealed with tape as specified for slotted style boxes in the appendix to the box specification.

5.1.2 Level B. Each tool with couplings shall be preserved and packaged as specified in 5.1.1 for level A except the box and the inserts, pads, or cushions used shall conform to PPP-B-636, type CF, class domestic, variety SW, grade 175, style optional. Box closure shall be in accordance with the appendix to the box specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

5.1.3 Level C. Each tool with couplings shall be preserved and packaged to afford protection against deterioration and damage from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Tools preserved and packaged as specified in 5.1, shall be packed together in a close fitting box conforming to PPP-B-601, overseas type, style optional in quantities not to exceed the weight limitation of the box. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall be zinc coated.

5.2.2 Level B. Tools preserved and packaged as specified in 5.1, shall be packed together as specified in 5.2.1 except boxes shall be domestic type and strapping need not be zinc coated. As an alternate the tools may be packed in a close fitting box conforming to PPP-B-640, class 2, style optional or PPP-B-636, V3c, V11c, V13c, or V15c, style optional. The gross weight or size of each box shall not exceed the limitations of the applicable box specification. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.3 Level C. Tools preserved and packaged as specified in 5.1, shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in a container complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The horizontal connecting tool is to be used to horizontally aline the deck sections of the class 60 floating bridge.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:15Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) Level of preservation and packaging and level of packing required (see 5.1 and 5.2). Level B preservation and packaging is intended to provide economical but limited protection and should be specified only when it has been determined that the tools will be held in covered storage.

6.3 Preproduction model. Any changes or deviations of production horizontal connecting tools from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish horizontal connecting tools conforming to this specification.

Custodian:

Army - ME

Preparing activity:

Army - ME

Project No. 5420-A182

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-52338A(ME) Tool, Horizontal Connecting: Erection Equipment for Bridge, Floating, Pneumatic Float, Class 60

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:15Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. ARMY MOBILITY EQUIPMENT
RESEARCH, DEVELOPMENT
CENTER
FORT BELVOIR, VIRGINIA 22060

OFFICIAL BUSINESS

AMSME-RZD-S

POSTAGE AND FEES PAID
DOD-314

COMMANDER
U. S. ARMY MOBILITY EQUIPMENT
RESEARCH, DEVELOPMENT CENTER
ATTN: AMSME-RZD-S
FORT BELVOIR, VIRGINIA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:15Z
Check the source to verify that this is the current version before use.